Week 6 — Feature Expansion Output

Goal

Add a second method/case/type and allow the user to choose it in the UI.

Required Outputs

☐ Second method/case/type implemented within project scope

☐ UI includes method selection (dropdown/radio/buttons)

☐ At least 5 total test cases (including new feature)

☐ Trail identifies which method was used

Evidence to Attach

☐ Screenshot: method selection control in UI

☐ Screenshot: trail showing chosen method

Reflection (Write short answers)

- How did your code structure change after adding the second feature?

Weekly Sign-off

Group Leader Name/Signature: _____ Date: _____

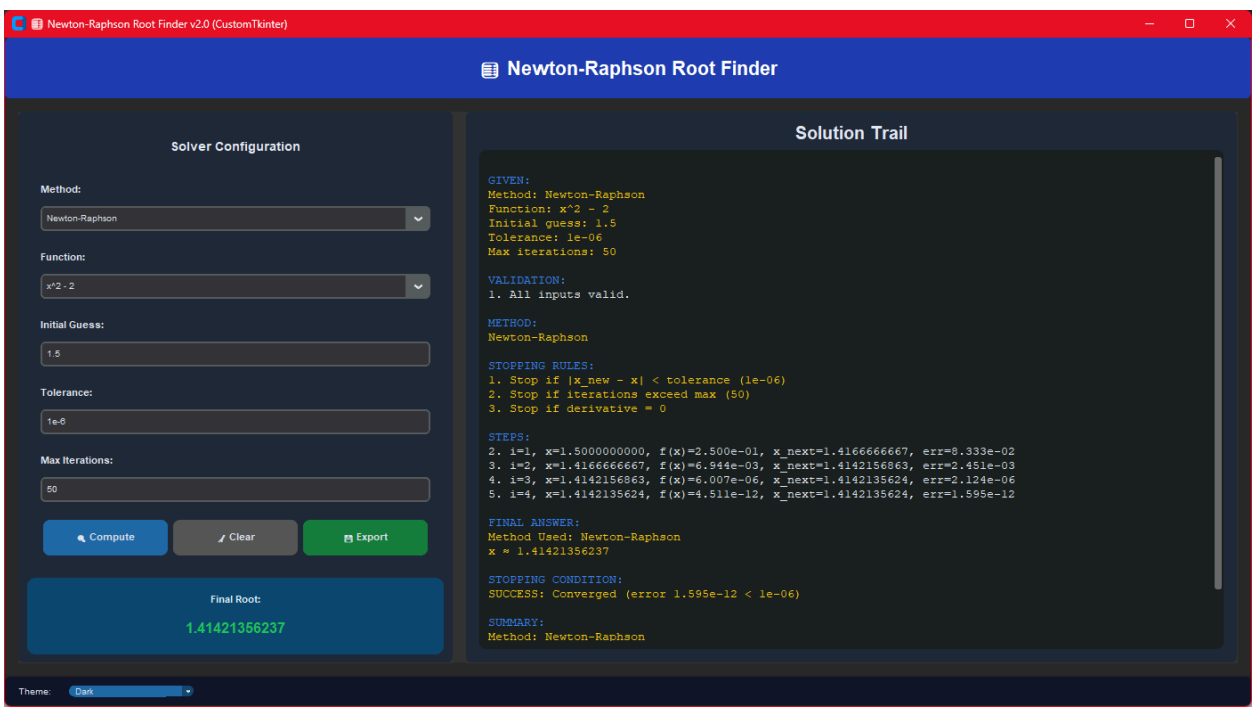
Member Name/Signature: _____ Date: _____

Member Name/Signature: _____ Date: _____

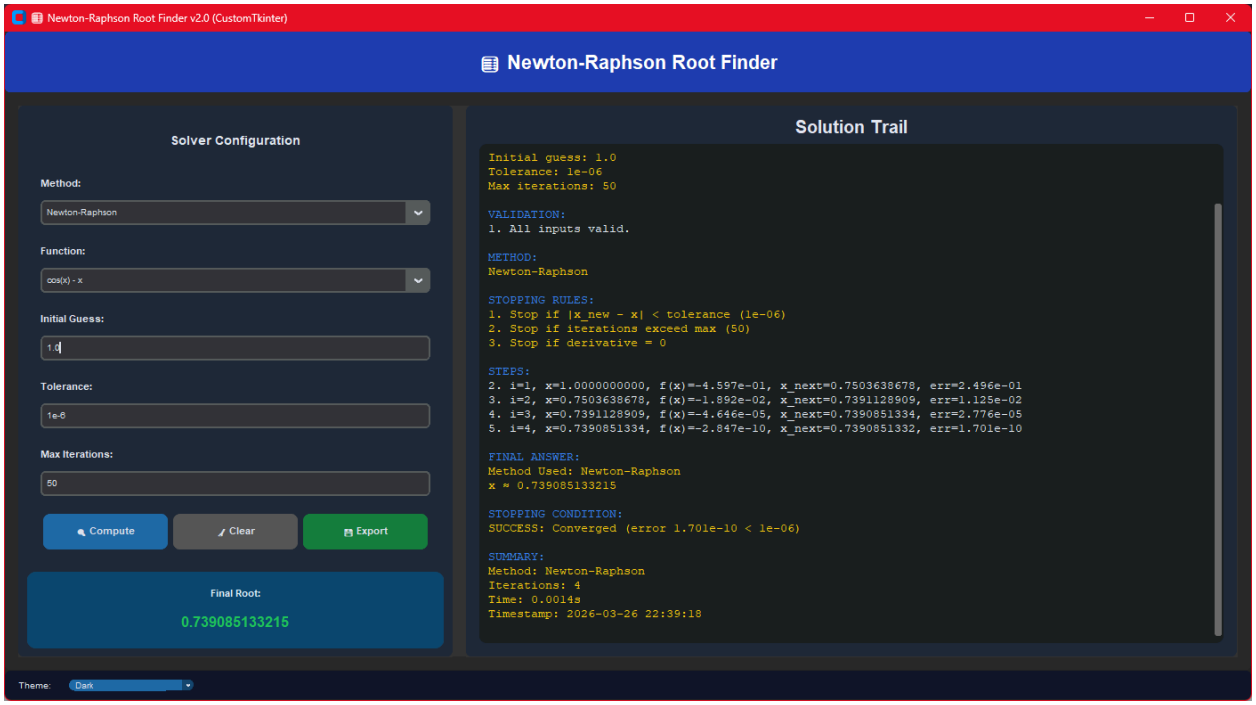
Test Cases

#	Function	Method	Initial Guess(es)	Expected Root	Status
1	$x^2 - 2$	Newton-Raphson	$x_0 = 1.5$	≈ 1.4142	☑
2	$\cos(x) - x$	Newton-Raphson	$x_0 = 1$	≈ 0.7391	☑
3	$x^3 - 3x + 1$	Newton-Raphson	$x_0 = 0.5$	≈ 0.3473	☑
4	$x^2 - 2$	Secant	$x_0 = 1, x_1 = 2$	≈ 1.4142	☑
5	$\cos(x) - x$	Secant	$x_0 = 0, x_1 = 1$	≈ 0.7391	☑

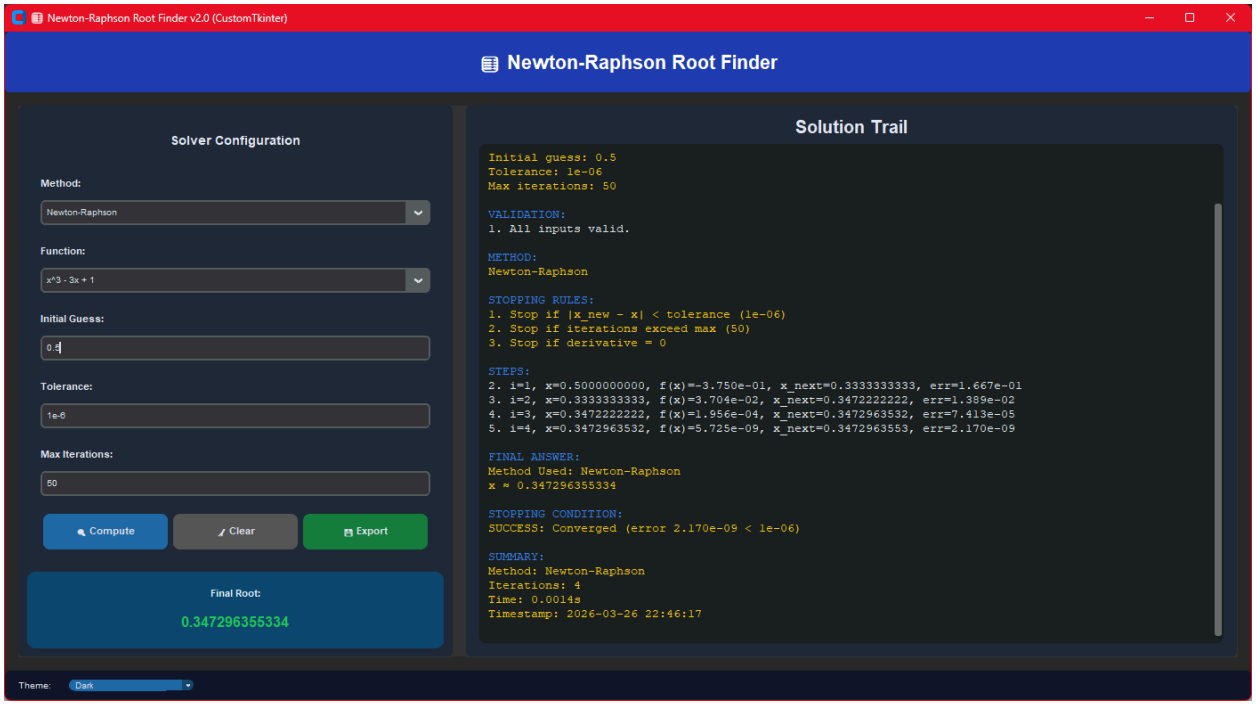
TEST #1



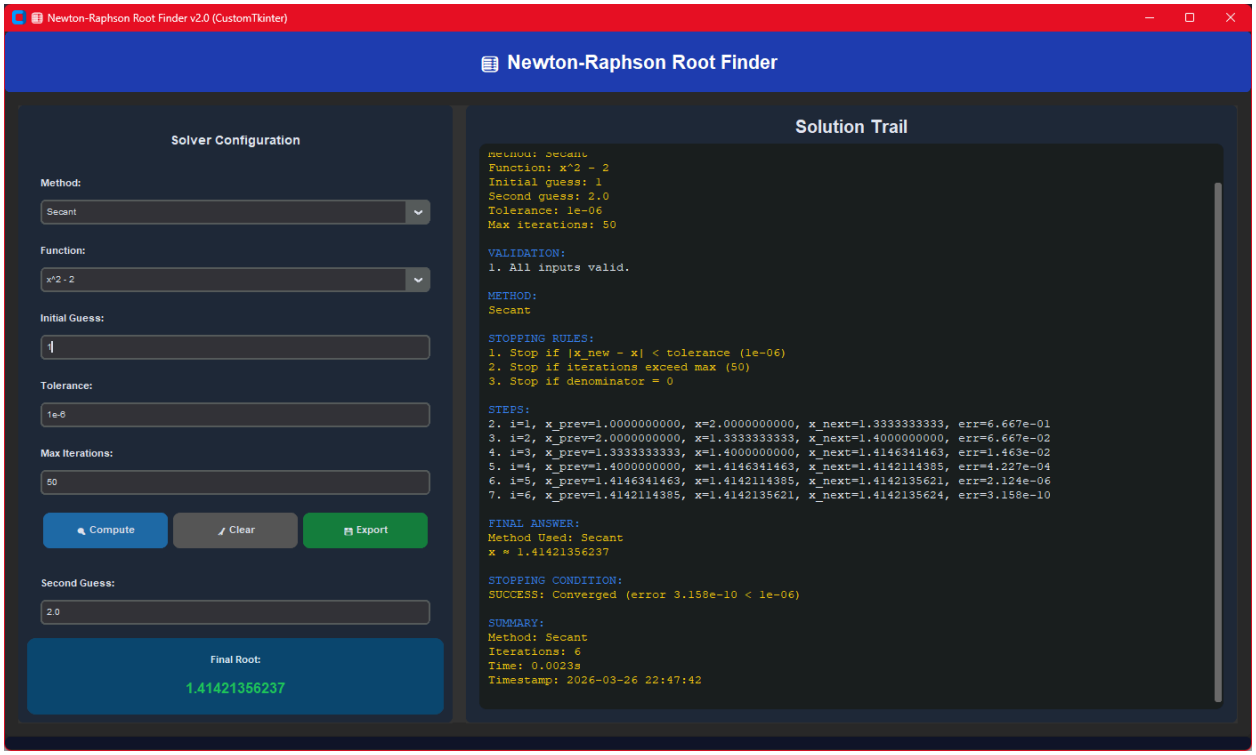
TEST#2


Newton-Raphson Root Finder v2.0 (CustomTkinter)
Newton-Raphson Root Finder
Solver Configuration
Method:
Newton-Raphson
Function:
cos(x) - x
Initial Guess:
1.0
Tolerance:
1e-6
Max Iterations:
50
Compute
Clear
Export
Final Root:
0.739085133215
Solution Trail
Initial guess: 1.0
Tolerance: 1e-06
Max iterations: 50
VALIDATION:
1. All inputs valid.
METHOD:
Newton-Raphson
STOPPING RULES:
1. Stop if |x_new - x| < tolerance (1e-06)
2. Stop if iterations exceed max (50)
3. Stop if derivative = 0
STEPS:
2. i=1, x=1.0000000000, f(x)=-4.597e-01, x_next=0.7503638678, err=2.496e-01
3. i=2, x=0.7503638678, f(x)=-1.892e-02, x_next=0.7391128909, err=1.125e-02
4. i=3, x=0.7391128909, f(x)=-4.646e-05, x_next=0.7390851334, err=2.776e-05
5. i=4, x=0.7390851334, f(x)=-2.847e-10, x_next=0.7390851332, err=1.701e-10
FINAL ANSWER:
Method Used: Newton-Raphson
x ≈ 0.739085133215
STOPPING CONDITION:
SUCCESS: Converged (error 1.701e-10 < 1e-06)
SUMMARY:
Method: Newton-Raphson
Iterations: 4
Time: 0.0014s
Timestamp: 2026-03-26 22:39:18
Theme:
Dark

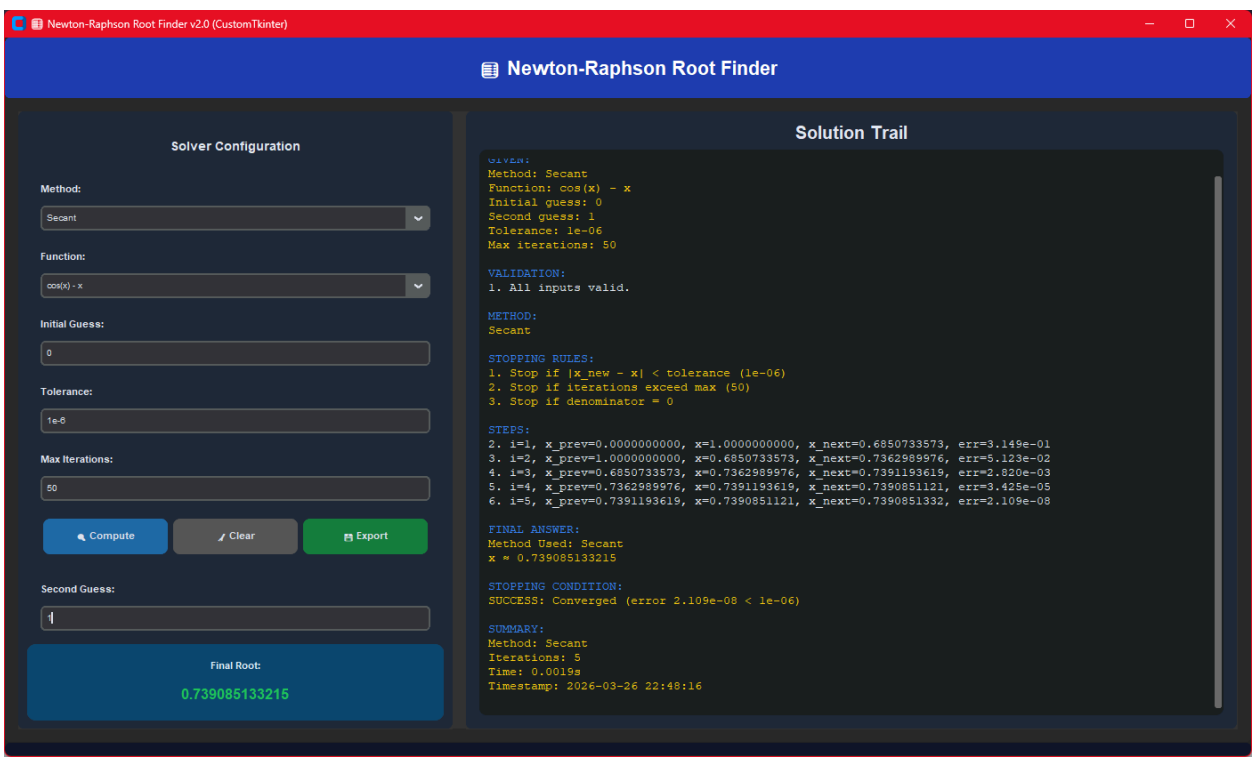
TEST#3


Newton-Raphson Root Finder v2.0 (CustomTkinter)
Newton-Raphson Root Finder
Solver Configuration
Method:
Newton-Raphson
Function:
x^3 - 3x + 1
Initial Guess:
0.5
Tolerance:
1e-6
Max Iterations:
50
Compute
Clear
Export
Final Root:
0.347296355334
Solution Trail
Initial guess: 0.5
Tolerance: 1e-06
Max iterations: 50
VALIDATION:
1. All inputs valid.
METHOD:
Newton-Raphson
STOPPING RULES:
1. Stop if |x_new - x| < tolerance (1e-06)
2. Stop if iterations exceed max (50)
3. Stop if derivative = 0
STEPS:
2. i=1, x=0.5000000000, f(x)=-3.750e-01, x_next=0.3333333333, err=1.667e-01
3. i=2, x=0.3333333333, f(x)=3.704e-02, x_next=0.3472222222, err=1.389e-02
4. i=3, x=0.3472222222, f(x)=1.956e-04, x_next=0.3472963532, err=7.413e-05
5. i=4, x=0.3472963532, f(x)=5.725e-09, x_next=0.3472963553, err=2.170e-09
FINAL ANSWER:
Method Used: Newton-Raphson
x ≈ 0.347296355334
STOPPING CONDITION:
SUCCESS: Converged (error 2.170e-09 < 1e-06)
SUMMARY:
Method: Newton-Raphson
Iterations: 4
Time: 0.0014s
Timestamp: 2026-03-26 22:46:17
Theme:
Dark

TEST#4

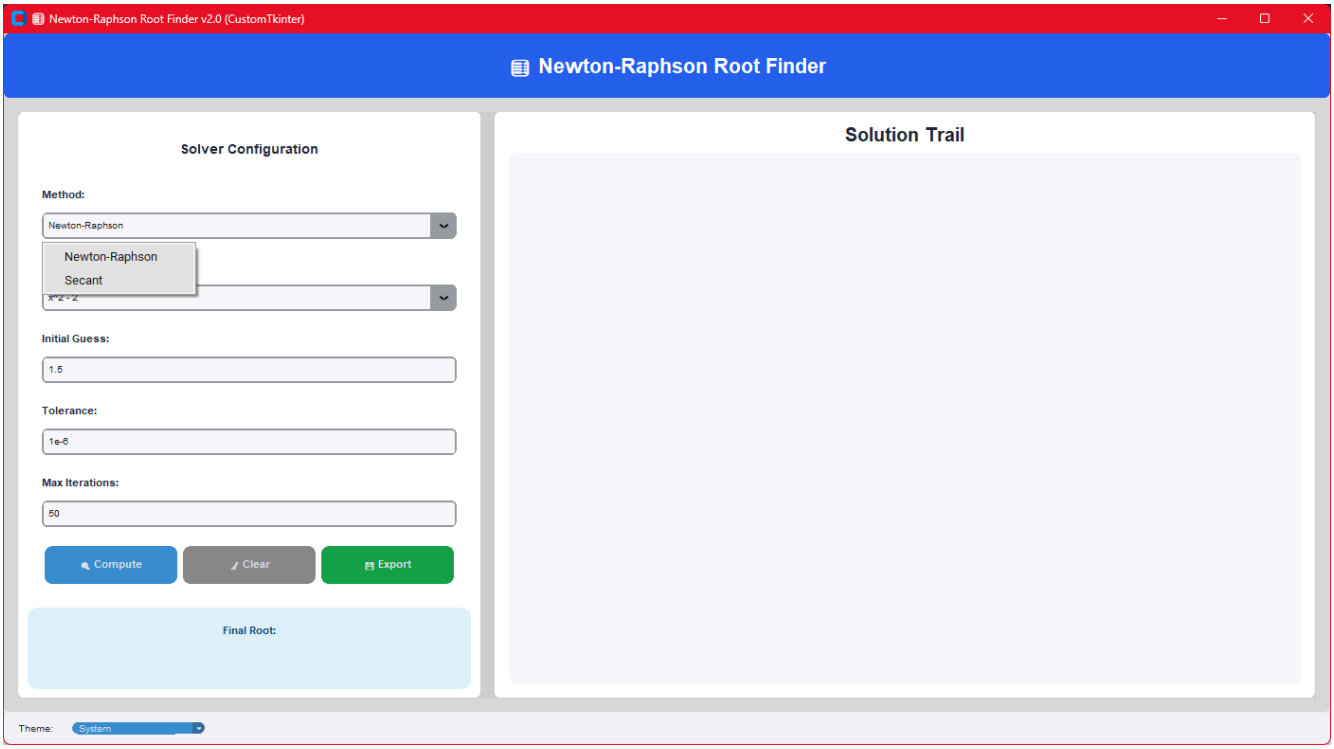

Newton-Raphson Root Finder v2.0 (CustomTkinter)
Newton-Raphson Root Finder
Solver Configuration
Method:
Secant
Function:
x^2 - 2
Initial Guess:
1
Tolerance:
1e-6
Max Iterations:
50
Compute
Clear
Export
Second Guess:
2.0
Final Root:
1.41421356237
Solution Trail
Function: x^2 - 2
Initial guess: 1
Second guess: 2.0
Tolerance: 1e-06
Max iterations: 50
VALIDATION:
1. All inputs valid.
METHOD:
Secant
STOPPING RULES:
1. Stop if |x_new - x| < tolerance (1e-06)
2. Stop if iterations exceed max (50)
3. Stop if denominator = 0
STEPS:
2. i=1, x_prev=1.0000000000, x=2.0000000000, x_next=1.3333333333, err=6.667e-01
3. i=2, x_prev=2.0000000000, x=1.3333333333, x_next=1.4000000000, err=6.667e-02
4. i=3, x_prev=1.3333333333, x=1.4000000000, x_next=1.4146341463, err=1.463e-02
5. i=4, x_prev=1.4000000000, x=1.4146341463, x_next=1.4142114385, err=4.227e-04
6. i=5, x_prev=1.4146341463, x=1.4142114385, x_next=1.4142135621, err=2.124e-06
7. i=6, x_prev=1.4142114385, x=1.4142135621, x_next=1.4142135624, err=3.158e-10
FINAL ANSWER:
Method Used: Secant
x ≈ 1.41421356237
STOPPING CONDITION:
SUCCESS: Converged (error 3.158e-10 < 1e-06)
SUMMARY:
Method: Secant
Iterations: 6
Time: 0.0023s
Timestamp: 2026-03-26 22:47:42

TEST#5


Newton-Raphson Root Finder v2.0 (CustomTkinter)
Newton-Raphson Root Finder
Solver Configuration
Method:
Secant
Function:
cos(x) - x
Initial Guess:
0
Tolerance:
1e-6
Max Iterations:
50
Compute
Clear
Export
Second Guess:
1
Final Root:
0.739085133215
Solution Trail
Method: Secant
Function: cos(x) - x
Initial guess: 0
Second guess: 1
Tolerance: 1e-06
Max iterations: 50
VALIDATION:
1. All inputs valid.
METHOD:
Secant
STOPPING RULES:
1. Stop if |x_new - x| < tolerance (1e-06)
2. Stop if iterations exceed max (50)
3. Stop if denominator = 0
STEPS:
2. i=1, x_prev=0.0000000000, x=1.0000000000, x_next=0.6850733573, err=3.149e-01
3. i=2, x_prev=1.0000000000, x=0.6850733573, x_next=0.7362989976, err=5.123e-02
4. i=3, x_prev=0.6850733573, x=0.7362989976, x_next=0.7391193619, err=2.820e-03
5. i=4, x_prev=0.7362989976, x=0.7391193619, x_next=0.7390851121, err=3.425e-05
6. i=5, x_prev=0.7391193619, x=0.7390851121, x_next=0.7390851332, err=2.109e-08
FINAL ANSWER:
Method Used: Secant
x ≈ 0.739085133215
STOPPING CONDITION:
SUCCESS: Converged (error 2.109e-08 < 1e-06)
SUMMARY:
Method: Secant
Iterations: 5
Time: 0.0019s
Timestamp: 2026-03-26 22:48:16

EVIDENCES

- [x] ~~Screenshot: method selection control in UI~~



- [x] ~~trail showing chosen method~~

```
GIVEN:
Method: Newton-Raphson
Function: x^2 - 2
Initial guess: 1.5
Tolerance: 1e-06
Max iterations: 50

VALIDATION:
1. All inputs valid.

METHOD:
Newton-Raphson

STOPPING RULES:
1. Stop if |x_new - x| < tolerance (1e-06)
2. Stop if iterations exceed max (50)
3. Stop if derivative = 0

STEPS:
2. i=1, x=1.5000000000, f(x)=2.500e-01, x_next=1.4166666667, err=8.333e-02
3. i=2, x=1.4166666667, f(x)=6.944e-03, x_next=1.4142156863, err=2.451e-03
4. i=3, x=1.4142156863, f(x)=6.007e-06, x_next=1.4142135624, err=2.124e-06
5. i=4, x=1.4142135624, f(x)=4.511e-12, x_next=1.4142135624, err=1.595e-12

FINAL ANSWER:
Method Used: Newton-Raphson
x ≈ 1.41421356237

STOPPING CONDITION:
SUCCESS: Converged (error 1.595e-12 < 1e-06)

SUMMARY:
Method: Newton-Raphson
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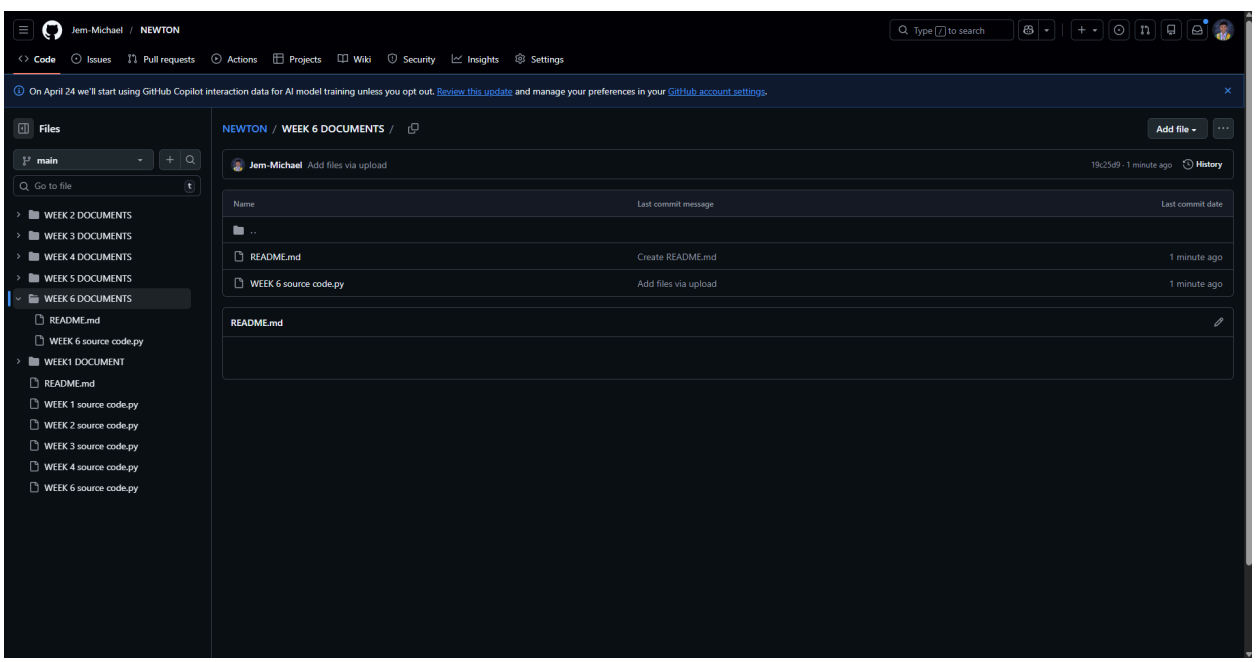
GITHUB REPO



Reflection

How did your code structure change after adding the second feature?

After adding the second method, the code structure changed a bit because it could no longer rely on just one fixed process. Before, the code was simpler since everything was based on a single method. When we added the second feature, we had to separate the logic for each method into different functions so they wouldn't get mixed up. We also added a way for the user to choose which method to use in the UI, so we needed a variable to store that choice and use it in the program flow. Because of this, the code became more organized and modular. It's now easier to understand and update, and it can add more methods in the future without changing too much of the existing code.

Weekly Sign-off

Group Leader Name/Signature: CHARLES RUSSEL O. SAN JUAN Date: 3/26/2026

Member Name/Signature: CHRISTINA GOMBA Date: 3/26/2026

Member Name/Signature: JOHN MICHAEL TULALIAN Date: 3/26/2026